Chapter 2.3: Checking data modelled by exponential distribution

Example 2.2

The R vector abbey in the package MASS gives 31 determinations of nickel content ($\mu g\, g^{-1}$) in a Canadian syenite rock.

```
> library(MASS)
> data(abbey)
> abbey
 [1]   5.2   6.5   6.9   7.0   7.0   7.0   7.4   8.0   8.0   8.0   8.0   8.5
[13]   9.0   9.0  10.0  11.0  11.0  12.0  12.0  13.7  14.0  14.0  14.0  16.0
[25]  17.0  17.0  18.0  24.0  28.0  34.0 125.0
>
```

We check whether the data are reasonably modelled by an exponential distribution.

There is no predefined function in R to construct an exponential Q-Q plot, so we have to work from first principles.

We start with the Exp(1) distribution and need to plot

$Q_e(p_i) = y_{(i)}$ against $Q(p_i)$, $i = 1, \ldots, 31$.

qexp(ppoints(31)) gives the theoretical quantile values at 31 probability points for the Exp(1) distribution.

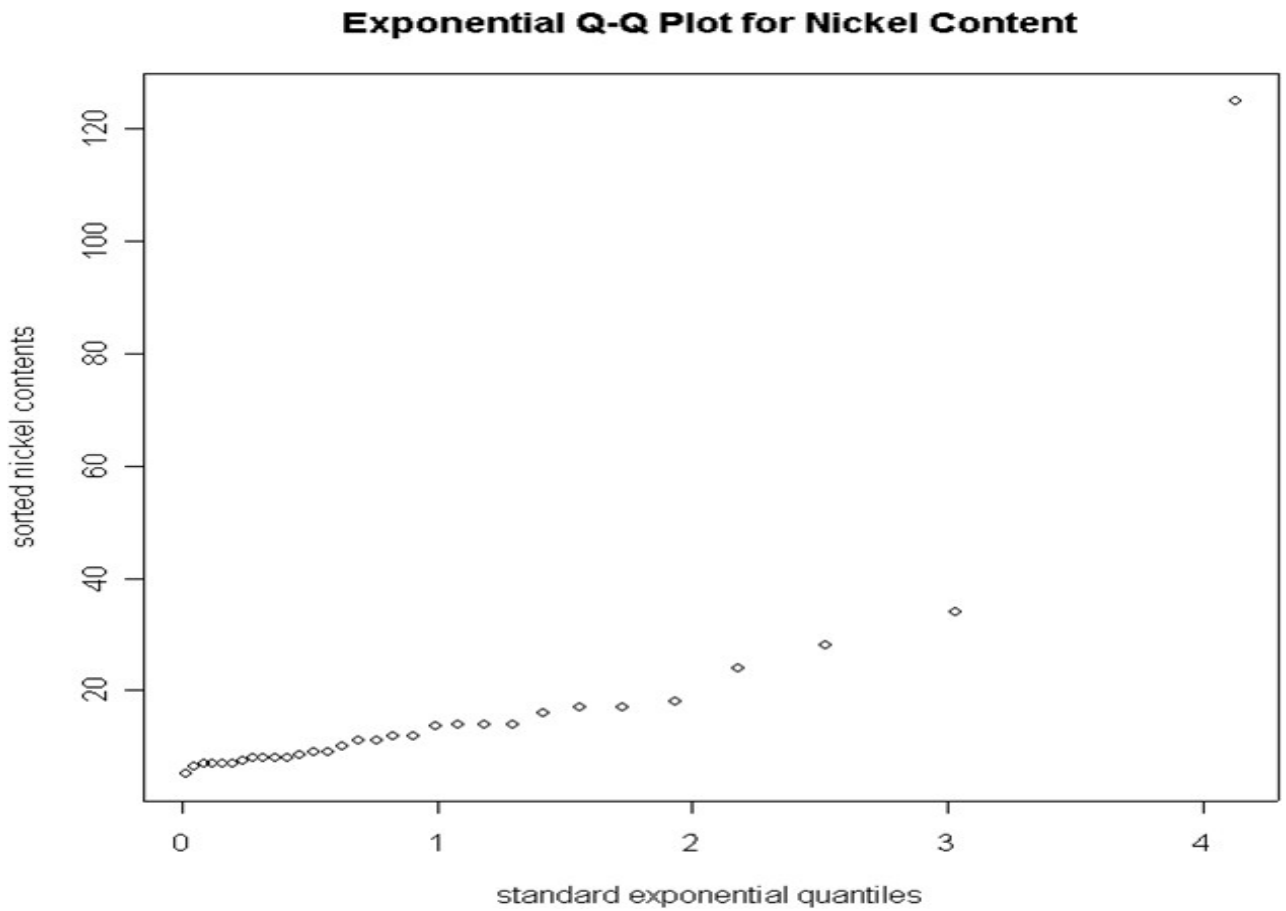
```
> qexp(ppoints(31))
 [1] 0.01626052 0.04959694 0.08408312 0.11980120 0.15684247 0.19530875
 [7] 0.23531409 0.27698678 0.32047190 0.36593427 0.41356232 0.46357274
[13] 0.51621647 0.57178632 0.63062682 0.69314718 0.75983856 0.83129752
[19] 0.90825856 0.99164017 1.08261195 1.18269541 1.29392104 1.41908418
[25] 1.56218503 1.72923911 1.92990981 2.18122424 2.51769647 3.02852210
[31] 4.12713439
>
```

sort(abbey) gives the sorted experimental values

```
> sort(abbey)
 [1]   5.2   6.5   6.9   7.0   7.0   7.0   7.4   8.0   8.0   8.0   8.0   8.5
[13]   9.0   9.0  10.0  11.0  11.0  12.0  12.0  13.7  14.0  14.0  14.0  16.0
[25]  17.0  17.0  18.0  24.0  28.0  34.0 125.0
>
```

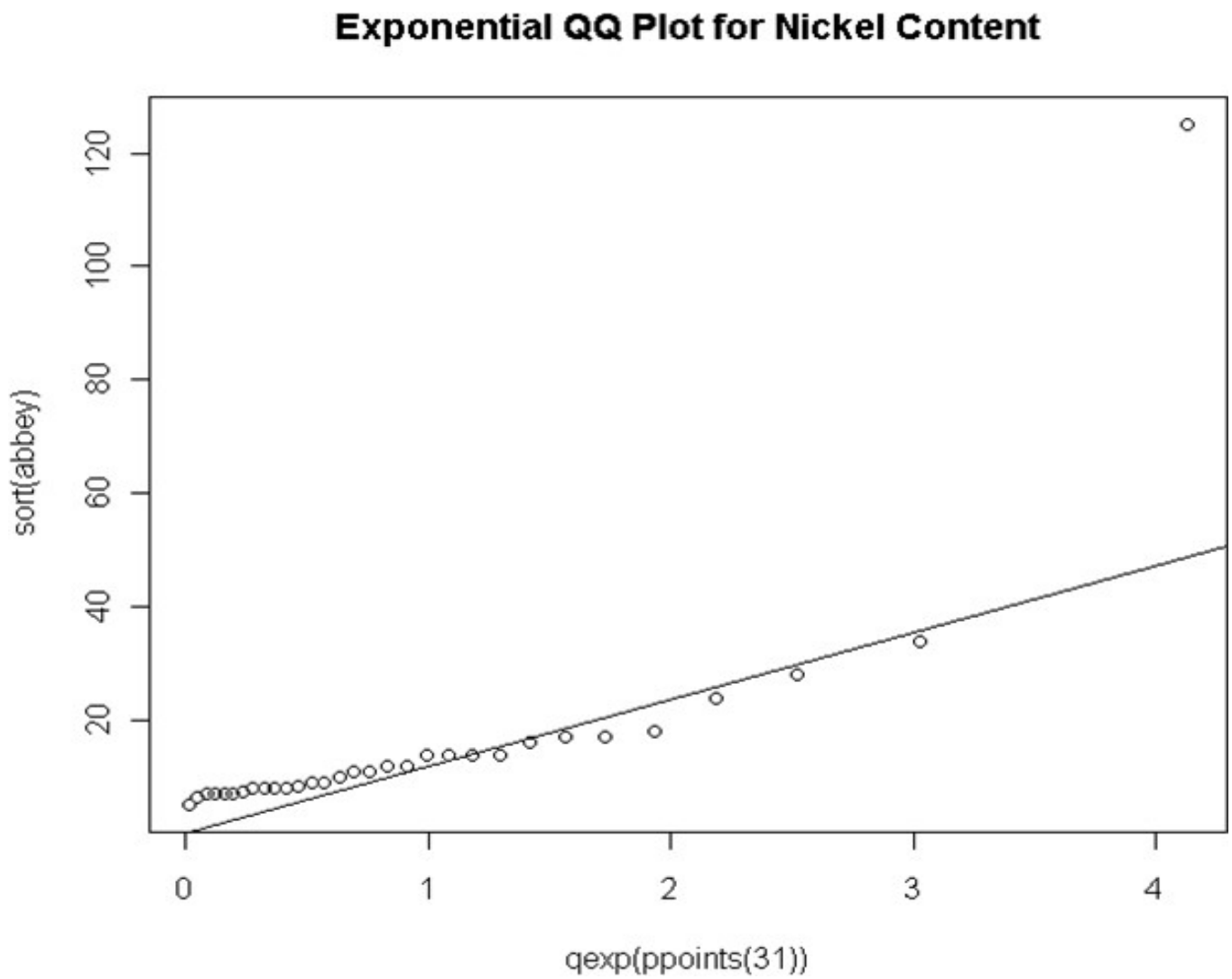
The following command produces a Q-Q plot with axes labelled accordingly.

```
Error: syntax error
> plot(qexp(ppoints(31)),sort(abbey),xlab="standard exponential quantiles",
+ ylab="sorted nickel contents",main="Exponential Q-Q Plot for Nickel Content")
>
```



Ignoring the extreme point, we can fit a straight line to the points (see later) and we find:

$$\hat{Q}_e(p) = 11.84\, Q(p)$$



So the original data (represented by r.v., say, X) seem to be described by the theoretical quantiles scaled by a factor of 11.84.

Therefore, the data follow the distribution of the transformed variable $Y = 11.84X$.

Since the QQ plot was produced assuming $X \sim \text{Exp}(1)$ we have

$$Y = 11.84X \sim \text{Exp}(1/11.84)$$

i.e. $\text{Exp}(0.0845)$.

If we ignore the highest observation, the data appear to be reasonably compatible with an $\text{Exp}(0.0845)$ distribution (mean = 11.84).

The line has been drawn using a *resistant regression* technique (more later).

We constrain it to go through the origin (hence the -1 in the code below)

```
> lqs(sort(abbey)~qexp(ppoints(31))-1)
Call:
lqs.formula(formula = sort(abbey) ~ qexp(ppoints(31)) - 1)

Coefficients:
qexp(ppoints(31))
            11.84

Scale estimates 4.853 3.529
```

Note that the line is then drawn on the plot that is already there using the abline command:

```
> plot(sort(abbey)~qexp(ppoints(31)))
> abline(0,11.84)
>
```

However, the probability that an Exp(0.0845) distribution would produce a value as extreme as the highest observation (125) is very small (2.6×10^{-5}).

So there must be considerable doubt about the suitability of an exponential model here.

A further possibility is that some transformation of the data may be modelled by a normal distribution.

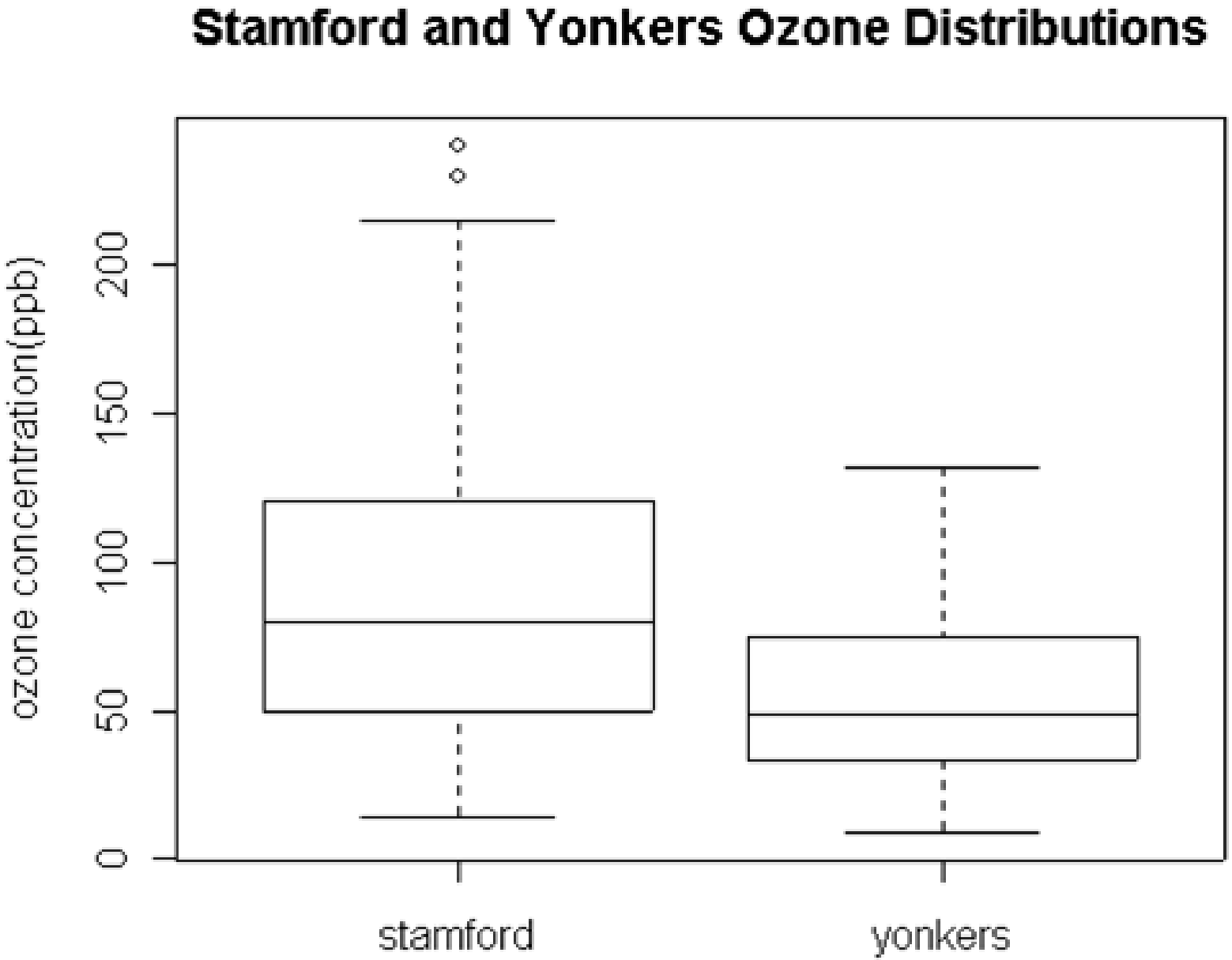
(*Try this at practicals*)

Boxplots

These are useful for comparing distributions of the same variable for different groups. The Rcode:

```
> boxplot(ozone,ylab="ozone concentration(ppb)",main="Stamford and Yonkers Ozone Distributions")
>
```

produces the following boxplot


Stamford and Yonkers Ozone Distributions
ozone concentration(ppb)
0
50
100
150
200
stamford
yonkers

Recall that the lower and upper ends of each box give the first and third quartiles of the corresponding distribution and the centre line indicates the median.